

Topics Covered:

- Spatial and Body Twists
- Summary
- Example

Additional Reading:

- LP 3.2.2 (Angular Velocity), 3.3.2 (Twists)
- MLS Chapter 2, Section 3.2 and Section 4

Spatial and Body Twists

Instead of always saying “Lie algebra element”, we call this a **twist**. In particular, we will use the notation ξ_b to denote the body twist and ξ_s to denote the spatial twist, with the following definitions:

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} \in \mathfrak{se}(2)/\mathfrak{se}(3) \\ \hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb} \in \mathfrak{se}(2)/\mathfrak{se}(3)\end{aligned}$$

Computing these gives us the individual formulas for body and spatial angular and linear velocity:

body twist:

$$\begin{aligned}\hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb} = g_{b\neq} \dot{g}_{\neq b} = \begin{bmatrix} R^\top & -R^\top d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{R} & \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} R^\top \dot{R} & R^\top \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_b]_\times & v_b \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(2)/\mathfrak{se}(3)\end{aligned}$$

spatial twist:

$$\begin{aligned}
 \hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} = \dot{g}_{sb} g_{bs} = \begin{bmatrix} \dot{R} & \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^\top & -R^\top d \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \dot{R} R^\top & -\dot{R} R^\top d + \dot{d} \\ 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} [\omega_s]_\times & v_s \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(2)/\mathfrak{se}(3)
 \end{aligned}$$

Note that in both cases, we can get the vector form by “unhatting”:

$$\xi_b = (\hat{\xi}_b)^\vee = \begin{Bmatrix} v_b \\ \omega_b \end{Bmatrix}, \quad \xi_s = (\hat{\xi}_s)^\vee = \begin{Bmatrix} v_s \\ \omega_s \end{Bmatrix}$$

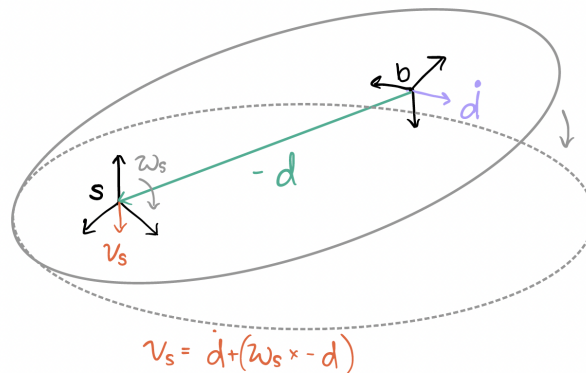
Also, note that in this formula, $v_s = -\dot{R}R^\top d + \dot{d}$ should NOT be interpreted as the linear velocity of the base-frame origin expressed in the fixed frame (that would simply be $\dot{d}$). Instead, the physical meaning is the instantaneous velocity of the body, currently at the fixed-frame origin, expressed in the fixed frame:

- ω_b : the angular velocity expressed in $\{b\}$
- ω_s : the angular velocity expressed in $\{s\}$
- v_b : the linear velocity of a point at the origin of $\{b\}$ expressed in $\{b\}$
- v_s : the linear velocity of a point at the origin of $\{s\}$ expressed in $\{s\}$

The intuition behind the expression for v_s can be derived as:

$$\begin{aligned}
 v_s &= \dot{d}_{sb} - \dot{R}R^\top d \\
 &= \dot{d}_{sb} - [\omega_s]_\times d \\
 &= \dot{d}_{sb} - \omega_s \times d \\
 &= \dot{d}_{sb} + \omega_s \times -d
 \end{aligned}$$

This is summarized by the diagram:



Lastly, we can change our reference frame from body to spatial by using adjoint transformation:

$$\xi_s = \text{Ad}_g \xi_b \quad (\text{Note here that } g \text{ is } g_{sb})$$

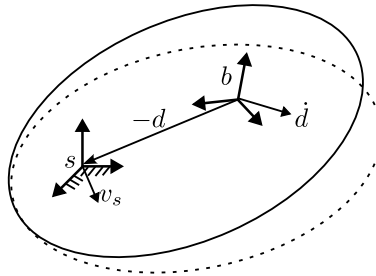
Where this adjoint is performed the same way as we learned earlier in the course:

$$\begin{aligned} \hat{\xi}_s &= g_{sb} \hat{\xi}_b g_{sb}^{-1} \\ &= \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} [\omega_b]_{\times} & v_b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^{\top} \dot{R} & R^{\top} \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} RR^{\top} \dot{R} & RR^{\top} \dot{d} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\top} & -R^{\top} d \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \dot{R} R^{\top} & -\dot{R} R^{\top} d + \dot{d} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_s]_{\times} & v_s \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Summary

Consider a rigid body moving in space, as shown in the following diagram:

- a) ω_b is the angular velocity expressed in the body frame, and ω_s is the angular velocity expressed in the fixed frame.
- b) v_b is the linear velocity of a point at the origin of b and expressed in b .
- c) v_s is the linear velocity of a point at the origin of s and expressed in s .



Definition: Spatial vs. Body Angular Velocity

Let $R(t)$ be a rotation matrix as seen from some fixed frame, i.e., $R(t) = R_{sb}(t)$, where s is the fixed frame (typically taken to be the world frame) and b is the body frame (attached to the rigid body). Denote $w \in \mathbb{R}^3$ as the angular velocity of the rotating frame.

Spatial angular velocity is fixed-frame vector representation of w (i.e., the angular velocity expressed in the fixed-frame coordinates which is typically defined to be the world frame) and is defined as:

$$[\omega_s]_{\times} = \dot{R}R^{-1} = \dot{R}R^{\top} \in \mathfrak{so}(3)$$

Body angular velocity is body-frame vector representation of w (i.e., the angular velocity expressed in the body frame) and is defined as:

$$[\omega_b]_{\times} = R^{-1}\dot{R} = R^{\top}\dot{R} \in \mathfrak{so}(3)$$

It may be helpful to note that the product $\dot{R}_{sb}R_{sb}^{-1}$ (for spatial angular velocity) is independent of b and the product $R_{sb}^{-1}\dot{R}_{sb}$ (for body angular velocity) is independent of s .

It also might be important to note that ω_b is *not* the angular velocity relative to a moving body frame, but rather it is relative to a *stationary* frame that instantaneously coincides with moving body frame. Otherwise the angular body velocity would always be zero.

Definition: Spatial vs Body Linear Velocity

Spatial linear velocity is the instantaneous velocity of the point (if the rigid body was sufficiently large) located at the fixed-frame origin, expressed in the fixed frame. It is defined as:

$$v_s = \dot{d} - \omega_s \times d = \dot{d} - \dot{R}R^{\top}d$$

Note: v_s is *not* the linear velocity of the body-fixed frame origin expressed in the fixed frame that quantity would simply be $\dot{d}$.

Body linear velocity is the velocity of a rigid body with respect to the (instantaneous) body frame. It is defined as:

$$v_b = R_{sb}^{-1}\dot{d} = R^{\top}\dot{d}$$

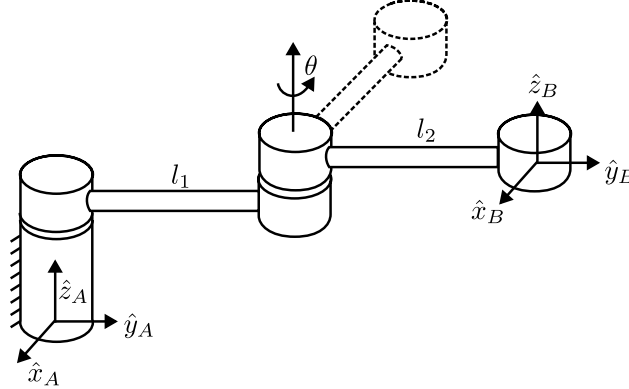
Derivations for these terms can be obtained through our derivation of the spatial and body

twists (note, this was done in previous lectures):

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb} g_{sb}^{-1} \\ \hat{\xi}_b &= g_{sb}^{-1} \dot{g}_{sb}\end{aligned}$$

Example

(Example 2.5 from MLS Chapter 2.4)



We can derive the configuration of this robot as:

$$\begin{aligned}
 g(t) &= g_{at} g_{tb} \quad (\text{Note: } t \text{ represents the temporary frame at the joint location}) \\
 &= \begin{bmatrix} R(\theta(t)) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R(0) & \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} R(\theta(t))R(0) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} + R(\theta(t)) \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} R(\theta(t)) & \begin{bmatrix} 0 \\ l_1 \end{bmatrix} + \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} 0 \\ l_2 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) & -l_2 \sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) & l_1 + l_2 \cos(\theta(t)) \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

The spatial velocity of the rotating rigid body can be solved for as:

$$\xi_s = \begin{Bmatrix} v_s \\ \omega_s \end{Bmatrix} = \begin{Bmatrix} -\dot{R}R^\top p + \dot{p} \\ (\dot{R}R^\top)^\vee \end{Bmatrix}$$

where p denotes the position of the body frame origin in the fixed frame:

$$p(t) = \begin{bmatrix} -l_2 \sin(\theta(t)) \\ l_1 + l_2 \cos(\theta(t)) \end{bmatrix}$$

and R is the orientation of the body frame origin in the fixed frame:

$$R(t) = \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix}$$

If we solve for these expressions you should get:

$$v_s = \begin{Bmatrix} l_1 \dot{\theta} \\ 0 \end{Bmatrix}, \quad \omega_s = \dot{\theta}$$

This computation was derived in a previous class using the homogeneous representation:

$$\begin{aligned} \hat{\xi}_s &= \dot{g}g^{-1} \\ &= \begin{bmatrix} \dot{R}R^\top & -\dot{R}R^\top p + \dot{p} \\ 0 & 0 \end{bmatrix} \quad (\text{Note that the form of this matrix is } \begin{bmatrix} [\omega_s]_\times & v_s \\ 0 & 0 \end{bmatrix}) \\ &= \begin{bmatrix} 0 & -\dot{\theta} & l_1 \dot{\theta} \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

We can see how we got this by solving for the terms:

$$\begin{aligned} \dot{R}R^\top &= \begin{bmatrix} -\sin(\theta(t))\dot{\theta} & -\cos(\theta(t))\dot{\theta} \\ \cos(\theta(t))\dot{\theta} & -\sin(\theta(t))\dot{\theta} \end{bmatrix} \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} = \begin{bmatrix} 0 & -\dot{\theta} \\ \dot{\theta} & 0 \end{bmatrix} \\ -\dot{R}R^\top p + \dot{p} &= \begin{bmatrix} 0 & \dot{\theta} \\ -\dot{\theta} & 0 \end{bmatrix} \begin{bmatrix} -l_2 \sin(\theta(t)) \\ l_1 + l_2 \cos(\theta(t)) \end{bmatrix} + \begin{bmatrix} -l_2 \cos(\theta(t))\dot{\theta} \\ -l_2 \sin(\theta(t))\dot{\theta} \end{bmatrix} \\ &= \begin{bmatrix} l_1 \dot{\theta} + \cancel{l_2 \cos(\theta(t))\dot{\theta}} - \cancel{l_2 \cos(\theta(t))(\dot{\theta})} \\ \cancel{l_2 \sin(\theta(t))\dot{\theta}} - \cancel{l_2 \sin(\theta(t))\dot{\theta}} \end{bmatrix} = \begin{bmatrix} l_1 \dot{\theta} \\ 0 \end{bmatrix} \end{aligned}$$

Again, note that the interpretation behind spatial velocity is that v_s is the velocity of a point attached to the rigid body as it travels through the origin of the A coordinate frame.

To solve for the body velocity, we follow a similar procedure:

$$\xi_b = \begin{Bmatrix} v_b \\ \omega_b \end{Bmatrix} = \begin{Bmatrix} R^\top \dot{p} \\ (R^\top \dot{R})^\vee \end{Bmatrix}$$

If we solve for these expressions you should get:

$$v_b = \begin{Bmatrix} -l_2 \dot{\theta} \\ 0 \end{Bmatrix}, \quad \omega_b = \dot{\theta}$$

As before, this computation can also be derived using homogeneous representation:

$$\begin{aligned} \hat{\xi}_b &= g^{-1} \dot{g} \\ &= \begin{bmatrix} R^\top \dot{R} & R^\top \dot{p} \\ 0 & 0 \end{bmatrix} \quad (\text{Note that the form of this matrix is } \begin{bmatrix} [\omega_b]_\times & v_b \\ 0 & 0 \end{bmatrix}) \\ &= \begin{bmatrix} 0 & -\dot{\theta} & -l_2 \dot{\theta} \\ \dot{\theta} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Again, we can check our computations:

$$\begin{aligned} R^\top \dot{R} &= \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} -\sin(\theta(t))\dot{\theta} & -\cos(\theta(t))\dot{\theta} \\ \cos(\theta(t))\dot{\theta} & -\sin(\theta(t))\dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & -\dot{\theta} \\ \dot{\theta} & 0 \end{bmatrix} \\ R^\top \dot{p} &= \begin{bmatrix} \cos(\theta(t)) & \sin(\theta(t)) \\ -\sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \begin{bmatrix} -l_2 \cos(\theta(t))\dot{\theta} \\ -l_2 \sin(\theta(t))\dot{\theta} \end{bmatrix} = \begin{bmatrix} -l_2 \dot{\theta} \\ 0 \end{bmatrix} \end{aligned}$$

The body velocity should be interpreted by imagining the velocity of the origin of the B frame, as seen by the B coordinates, so linear velocity is always in the $-x$ direction and angular velocity is always in the z direction (for this problem!).